

Oct20

Determine whether the following sequence:

$$\sum_{n=1}^{\infty} \frac{n^2 + 2n + 1}{n^3}$$

Converges or diverges

$$\frac{n^2}{n^3} + \frac{2n}{n^3} + \frac{1}{n^3}$$

$\downarrow$
 $= \frac{1}{n}$
 $\sum_{n=1}^{\infty} \frac{1}{n}$
 diverges

$\downarrow$
 $\sum_{n=1}^{\infty} \frac{2}{n^2}$
 converges

$\downarrow$
 $\sum_{n=1}^{\infty} \frac{1}{n^3}$
 converges

p-series where p is > 1 , then converges
 SCRAP, illegal

We can only break things up if everything converges - You cannot recombine

$$\frac{n^2 + 2n + 1}{n^3} \geq \frac{1}{n^3} = \frac{1}{n}$$

$$\sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+2} \right) = 2$$

Diverges via comparison test - why?

$$\underbrace{\sum_{n=1}^{\infty} \frac{1}{n}}_{\text{Diverges}} \text{ vs } \sum_{n=1}^{\infty} \frac{1}{n+2}$$

Factorials

$$5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5$$

$$6! = 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 6 \cdot 5!$$

$$(n+2)! = (n+2) n!$$

important